

BECM 3107

LIQUIDATED DAMAGES (LD) & PENALTIES

Term Final Exam Study Guide

Easy explanations | সহজ ব্যাখ্যা

Memorizing techniques | মনে রাখার কৌশল

Bangla meanings | প্রয়োজনীয় বাংলা অর্থ

Exam questions, calculations and quick revision

Based on the uploaded BECM 3107 course slides

Liquidated Damage (LD) & Penalties - Dr. Mizanoor Rahman, P.Eng., Department of BECM, KUET

Focused revision companion for the term final

A How to Use This Guide

Read the rule, memorize the code, then answer without looking.

এই গাইডটি মূল 40-page slide deck-এর terminology, sequence এবং examples অনুসরণ করে তৈরি করা হয়েছে। সহজ বাংলা ব্যাখ্যা, mnemonic, vocabulary, short calculation, probable questions এবং revision plan দেওয়া আছে।

3R

Read -> Recall -> Review

প্রথমে বুঝে পড়ুন, পরে বই বন্ধ করে মনে করুন, শেষে ডুল অংশ revise করুন।

1. Read the English rule and the Bangla explanation together.
2. Memorize the code or one-line rule.
3. Write one short answer from memory.
4. Before the exam, revise only the master table, cases, and vocabulary.

B Course Map

Part	Topic	Main focus
1	LD and Penalty	Meaning, purpose, effect, enforceability
2	Standard Forms	JCT, NEC3, FIDIC and phasing
3	Quantum Meruit	Reasonable payment where contractual price entitlement fails
4	Time and Completion	Commencement, progress, completion, EOT
5	Post-completion	Testing, warranty, remedy and guaranty

Core idea: LD is an agreed pre-estimate of loss for breach such as delay; a penalty is primarily designed to punish. Enforceability depends on the clause and the required contractual procedure.

1 Construction Timeline and the LD Trigger

Understand where delay damages arise in the project sequence.

The project-progression diagram in the source places liquidated damages after the contractual date for completion, during the contractor-in-delay period, and before practical completion. The wider sequence includes possession, contract period, extension of time, completion, rectification, making-good certification, and final certification.

Stage	Meaning	Why it matters for LD
Date of possession	Employer gives access to the site	The contract period normally begins from the agreed possession/commencement arrangement.
Date for completion	Contractual deadline	Delay is measured against this date or a validly revised date.
Extension of Time (EOT)	Approved adjustment of the completion date	Valid EOT moves the deadline and protects the LD mechanism.
Contractor in delay	Works remain incomplete after the applicable deadline	LD may accrue if the clause and procedures are enforceable.
Practical completion	Works are sufficiently complete for intended use	LD liability for late completion normally stops.
Rectification period	Defects are corrected after completion	This is distinct from the original delay period.

**P-D-E-
D-C**

Possession -> Deadline -> EOT -> Delay -> Completion

সাইট হস্তান্তর, শেষ তারিখ, সময় বৃদ্ধি, বিলম্ব, সমাপ্তি।

সহজ বাংলা: Contractor যদি নির্ধারিত completion date বা valid EOT-এর পরেও কাজ শেষ না করে, তখন enforceable LD clause অনুযায়ী প্রতিদিন/প্রতি সপ্তাহে agreed amount কাটা হতে পারে।

Possible exam question: Describe the key events in construction project progression relevant to LD.

- Start from possession and the contract period.
- Explain the date for completion and extension of time.
- Identify the contractor-in-delay period and practical completion.
- Mention rectification and final certification as later stages.

2 Liquidated Damages and Penalty Damages

This distinction is the foundation of the chapter.

Point	Liquidated Damages (LD)	Penalty damages
Meaning	An amount agreed in advance as compensation for breach, such as delay.	An amount set higher than the actual loss.
Purpose	Compensate the employer for costs/loss caused by late completion.	Punish or deter the party in breach, not merely compensate.
Relationship to loss	A genuine covenanted pre-estimate of damage.	May be extravagant or disproportionate to actual loss.
Common-law position in slides	Enforceable if it is genuine and procedures are followed.	Usually not enforced by common-law courts.
Example	\$100 per delayed day agreed in the contract.	\$50,000 penalty for delay even where actual loss is much lower.

C-P

Compensation = LD; Punishment = Penalty

LD ক্ষতিপূরণমূলক, penalty শাস্তিমূলক।

Exam sentence: An LD clause fixes compensation in advance; a penalty clause imposes a sum whose dominant purpose is punishment.

Practice calculation: If the agreed LD is \$100 per day and the enforceable delay is 18 days, the basic amount is $\$100 \times 18 = \$1,800$, subject to any contractual maximum and the validity of the clause.

Possible exam question: Distinguish liquidated damages from penalty damages.

- Define each term.
- Compare purpose, relationship to actual loss, and enforceability.
- Use a delay-rate example.
- Conclude that the court looks at substance, not merely the label used by the parties.

3 Purpose and Effect of an LD Clause

The clause benefits both employer and contractor by creating certainty.

The slides present the LD clause as mutually beneficial. The parties agree beforehand on damages recoverable for breach and can assess their liabilities and risks with reasonable certainty.

Employer benefit	Contractor benefit	Shared benefit
Knows the extent of protection if the contractor fails to perform.	Exposure to the employer's delay claim is limited to the agreed regime.	Both parties can price and manage risk before entering the contract.

Effect of the clause:

- Once the specified breach is triggered, the agreed sum generally caps the amount recoverable.
- Loss over and above the stipulated amount is generally not recoverable under that LD regime.
- Example from the slides: if the clause fixes \$10,000, the employer normally cannot recover \$15,000 merely because actual loss is higher.

C-C-L

Certainty -> Cap -> Limited exposure

আগে থেকে নিশ্চিত অঙ্ক, recovery cap, contractor-এর সীমিত exposure।

Important: A cap is useful only if the LD clause remains enforceable. A penalty, unmet condition precedent, or time-at-large problem may defeat the clause.

4 Nature and Enforceability of LD

A genuine pre-estimate can be enforced; a penalty cannot.

Liquidated damages are described as a genuine covenanted or agreed pre-estimate of damage. If the agreed sum is legally regarded as a penalty, the courts will not enforce it.

Case **Dunlop Pneumatic Tyre Co. v New Garage & Motor Co. Ltd**

The slides cite Dunlop as the principal case on distinguishing a genuine liquidated-damages provision from an unenforceable penalty. For this course answer, connect the case name to the genuine-pre-estimate versus penalty distinction.

G-P-E

Genuine pre-estimate = Enforceable; Penalty = Unenforceable

প্রকৃত পূর্বানুমান enforceable; শাস্তিমূলক অঙ্ক enforceable নয়।

Sectional work problem: a single LD sum may fail where the works are to be completed in sections or the employer takes possession of part before the whole is completed. A clause should match the sectional obligations rather than impose one indiscriminate figure.

সহজ উদাহরণ: তিনটি section-এর project-এ দুইটি section সময়মতো শেষ ও ব্যবহার শুরু হয়েছে, কিন্তু একটি section late। পুরো project-এর জন্য একক full-rate LD অযৌক্তিক বা unenforceable হতে পারে।

Possible exam question: Explain the nature of the contractual agreement for LD.

- State genuine covenanted pre-estimate of damage.
- Explain that a penalty is not enforced.
- Mention Dunlop.
- Discuss why a single sum may fail for sectional or partially possessed works.

5 Defences Against Enforcement of LD

The contractor can challenge the clause or the employer's procedure.

Defence / problem	Rule	Example or result
Penalty	The stipulated sum is punitive rather than a genuine pre-estimate.	The court does not enforce the amount.
Condition precedent not fulfilled	A required certificate or notice must be issued before LD liability arises.	No Architect's Delay Certificate or required notice.
Employer prevention	Employer conduct delays completion.	The contract must provide an effective EOT mechanism.
Time at large	The prevention event is not covered by the EOT clause.	Contractor must finish within a reasonable time; LD becomes inoperable.
Sectional/partial possession mismatch	Full LD cannot fairly apply after part is taken over.	Rate should be reduced or apportioned under the contract.

P-C-T-S

Penalty - Condition - Time at large - Sections

চারটি প্রধান challenge: penalty, condition precedent, time at large, sectional issue।

Time at large: When an employer-caused prevention event is not accommodated by the EOT clause, the fixed completion date may cease to control. The contractor then completes within a reasonable time, and the LD clause cannot operate.

Possible exam question: What defences may be raised against enforcement of an LD clause?

- Penalty character.
- Failure of a condition precedent such as certificate or notice.
- Employer prevention and ineffective EOT machinery.
- Time at large.
- Sectional completion or partial possession.

6 LD Clauses in Standard Forms

Different forms use different procedural machinery.

The slides list JCT, MasterFormat/CSI, FIDIC, NEC, SIA, ACA and PSSCOC as examples of standard conditions of contract. The detailed LD discussion focuses on JCT, NEC3 and FIDIC.

Form / clause	Procedure or rule in the slides
JCT SBC 11 cl. 2.32	Before claiming or deducting LD, the contract administrator issues a non-completion certificate and the employer gives written notice of intention to deduct.
JCT after EOT	Each newly fixed completion date requires a fresh non-completion certificate; the original notice remains effective unless revoked.
NEC3 cl. X7	LD becomes payable/deductible when completion is late beyond prescribed time or valid extension; no specific engineer's non-completion certificate is required.
JCT cl. 2.37 / NEC3 X7.3 / FIDIC 1999 Red Book cl. 10.2	After partial possession/taking over, the delay-damages rate is reduced in proportion to the value of the part taken over.

J-N-P

JCT = Notice/certificate; NEC = prescribed time; Phasing = proportionate reduction

JCT procedure, NEC trigger, partial possession-এ proportionate reduction।

Exam caution: Do not mix the procedures. JCT requires the stated certificate and notice; the slide says NEC3 clause X7 does not specifically require certification of non-completion.

Illustration from the source table: specific works may have separate completion periods and separate daily LD rates, such as \$10,000 per day for identified areas. This supports carefully drafted sectional obligations.

7 Quantum Meruit Claims

Payment of the reasonable value of work done.

A quantum meruit claim allows a contractor to seek the reasonable value of work performed where there is no contractual entitlement to payment or no contractual method for assessing the amount due.

Bangla meaning: Quantum meruit = কাজের যুক্তিসঙ্গত মূল্য অনুযায়ী পাওনা। নির্ধারিত contractual price না থাকলে বা price mechanism কাজ না করলে already performed work-এর reasonable value দাবি করা হয়।

Situation from slides	Why quantum meruit may arise
Express undertaking to pay a reasonable sum	Employer promised reasonable payment for services.
Services requested under a letter of intent, but no price agreed	A reasonable sum may be implied.
Price-fixing clause fails to operate	Contract gives no workable assessment of amount due.
Extra work lies outside the variations clause	No contractual pricing entitlement covers the extra work.
Apparent contract is void	Work was done but the supposed contract is legally ineffective.
Contract cancelled before completion	Contractor may seek reasonable value for work already completed.

N-P-R

No price / Price mechanism fails -> Reasonable value

দাম নেই বা price clause ব্যর্থ -> reasonable payment।

Possible exam question: Define quantum meruit and state the situations in which it arises.

- Define it as reasonable value of work done.
- Explain absence of contractual entitlement or assessment.
- List letter of intent/no agreed price, failed price clause, extra work, and void contract.
- Give the cancelled-house-work example.

8 Time Framework: Commencement and Progress

Time obligations begin with possession/commencement and continue through steady progress.

The slides divide time into commencement, progress, completion, post-completion obligations and extension of time. Commencement includes possession of the site, the date for possession and the move toward completion.

Progress under FIDIC Red Book clause 8.1: the contractor must proceed with due expedition and without delay. The contractor should work continuously, industriously and efficiently, using appropriate physical resources and progressing substantially in accordance with time, sequence and quality requirements.

C-I-E

Continuous - Industrious - Efficient

কাজ চলমান, পরিশ্রমী ও দক্ষভাবে এগোতে হবে।

Term	Easy meaning	বাংলা অর্থ
Due expedition	Proceed promptly and properly	যথাযথ দ্রুততায়
Without delay	Avoid unjustified delay	অযৌক্তিক বিলম্ব ছাড়া
Physical resources	Labour, plant, materials and site resources	শ্রমিক, যন্ত্রপাতি, উপকরণ ও সম্পদ
Sequence	Required order of work activities	কাজের ধারাবাহিক ক্রম

9 Completion: Meaning, Date and Effects

Completion is both a physical achievement and a contractual certification event.

Under FIDIC Red Book 1999 clause 8.2, the contractor must complete the whole works, pass tests on completion, and finish work necessary for taking over. Clauses 9.1-9.4 deal with testing. A Taking-Over Certificate may be issued despite minor outstanding work or defects that do not substantially affect intended use.

Date for completion: a contract need not specify a date, but without one the contractor must complete within a reasonable time. The slides stress that this is commercially unsatisfactory for major projects and prevents the use of a fixed LD regime.

Effect of completion certification	Result
Possession	Employer becomes entitled to take possession of the works.
Insurance responsibility	Contractor's responsibility to insure completed works generally ends, subject to terms.
LD	Liability for late-completion LD usually ceases.
Retention	Part, often one-half, may be released depending on contract conditions.

P-I-L-R

Possession - Insurance - LD stops - Retention release

সমাপ্তির চার ফল: possession, insurance, LD stop, retention release।

Possible exam question: Explain the meaning and effects of completion.

- Mention whole works, tests and taking over.
- Allow for minor defects not substantially affecting use.
- Explain why a completion date matters.
- List possession, insurance, cessation of LD and retention release.

10 After Completion and Adjustment of Time

Post-completion testing and EOT rules protect the contractual time mechanism.

The slides refer to FIDIC Red Book clauses 11.1-11.11 for post-completion procedures. If works fail tests on completion, further tests may be required. After repeated failures, clause 9.4 may permit a Taking-Over Certificate, with a proportionate reduction of the contract price and possible independent completion of remaining work.

Adjustment of time: the completion time can be adjusted only where the contract permits and strictly according to the contractual procedure. If an event is not covered, the contractor cannot demand more time and the employer cannot simply grant it outside the agreed mechanism.

Ground / source item	Time effect described in slides
Variation - JCT SBC 11 cl. 2.29	A variation may entitle the contractor to an extension of time.
Contract administrator instruction	Some instructions causing delay are relevant events under cl. 2.29.2.
Employer prevention	A workable EOT clause preserves the employer's LD right.
Uncovered prevention event	May make time at large and LD inoperable.

P-P-P

Permission - Procedure - Proof

EOT-এর জন্য contract permission, সঠিক procedure এবং delay proof দরকার।

Source limitation: The payment slide only says "Refer to John Murdoch page 229" and provides no further payment rules. This guide therefore does not add unsupported payment details.

11 Warranty, Remedy and Guaranty

Separate the promise of quality, damages for breach and post-completion correction duty.

Concept	Rule in the slides	বাংলা সহজ অর্থ
Warranty	Contractor represents/covenants that work will meet stated standards and materials will be free from defects.	কাজ ও উপকরণ নির্ধারিত মানের হবে - এই চুক্তিগত নিশ্চয়তা।
Remedy for breach	The other party may recover monetary damages, such as repair or replacement cost.	ত্রুটির কারণে repair/replacement-এর খরচ ক্ষতিপূরণ হিসেবে আদায়।
Guaranty / Guarantee	After completion, contractor must replace defective/non-conforming materials or remedy improper workmanship at own cost during the stated period.	নির্দিষ্ট guarantee period-এ নিজের খরচে ত্রুটি ঠিক বা উপকরণ বদলানো।

W-R-G

Warranty = promise; Remedy = money; Guaranty = correct defects

মানের প্রতিশ্রুতি, আর্থিক প্রতিকার, পরে ত্রুটি সংশোধন।

Possible exam question: Distinguish warranty, remedy and guaranty in a construction contract.

- Warranty states quality/performance standards.
- Breach may result in monetary damages.
- Guaranty operates after completion for a stated period and requires correction at contractor cost.

12 One-Page Master Revision

Use this table in the final hour before the exam.

Topic	Memory code	One-line rule
Project time	P-D-E-D-C	Possession, deadline, EOT, delay, completion.
LD vs penalty	C-P	LD compensates; penalty punishes.
Purpose/effect	C-C-L	Certainty, cap and limited exposure.
Enforceability	G-P-E	Genuine pre-estimate enforceable; penalty not.
Defences	P-C-T-S	Penalty, condition, time at large, sections.
Standard forms	J-N-P	JCT procedure, NEC trigger, proportionate phasing.
Quantum meruit	N-P-R	No price/failed mechanism -> reasonable value.
Progress	C-I-E	Continuous, industrious and efficient.
Completion effects	P-I-L-R	Possession, insurance, LD stops, retention.
EOT	P-P-P	Permission, procedure and proof.
Quality obligations	W-R-G	Warranty, remedy and guaranty.

LD calculation pattern: Applicable LD rate x enforceable delay period, subject to the stated maximum, EOT, completion certification, phasing and enforceability requirements.

Key authorities / clauses: Dunlop; JCT SBC 11 clauses 2.32, 2.37 and 2.29; NEC3 clauses X7 and X7.3; FIDIC 1999 Red Book clauses 8.1, 8.2, 9.1-9.4, 10.2 and 11.1-11.11.

13 Essential Vocabulary with Bangla Meanings

English	বাংলা অর্থ	English	বাংলা অর্থ
Liquidated damages	পূর্বনির্ধারিত ক্ষতিপূরণ	Penalty	শাস্তিমূলক অর্থদণ্ড
Employer	নিয়োগকারী / প্রকল্প মালিক	Contractor	ঠিকাদার
Late completion	বিলম্বিত সমাপ্তি	Breach of contract	চুক্তি লঙ্ঘন
Pre-estimate	আগাম ক্ষতির অনুমান	Covenanted	চুক্তিতে সম্মত
Enforceable	আইনগতভাবে কার্যকরযোগ্য	Inoperable	কার্যকর করা যায় না
Condition precedent	পূর্বশর্ত	Delay certificate	বিলম্ব সনদ
Non-completion certificate	অসমাপ্তির সনদ	Written notice	লিখিত নোটিশ
Extension of Time (EOT)	সময় বৃদ্ধির অনুমোদন	Act of prevention	কাজে বাধা সৃষ্টি করা ঘটনা
Time at large	নির্দিষ্ট completion date অকার্যকর হওয়া	Reasonable time	যুক্তিসঙ্গত সময়
Phasing	পর্যায়ভিত্তিক কাজ	Partial possession	আংশিক দখল/ব্যবহার গ্রহণ
Apportionment	আনুপাতিক ভাগ	Quantum meruit	কাজের যুক্তিসঙ্গত মূল্য
Letter of intent	কাজ শুরুর অভিপ্রায়পত্র	Variation	পরিবর্তিত/অতিরিক্ত কাজ
Commencement	কাজ শুরু	Due expedition	যথাযথ দ্রুততা
Practical completion	ব্যবহারোপযোগী বাস্তব সমাপ্তি	Taking-Over Certificate	কাজ গ্রহণের সনদ
Retention money	আটক রাখা অর্থ	Rectification period	ত্রুটি সংশোধনকাল
Warranty	মানের চুক্তিগত নিশ্চয়তা	Defect	ত্রুটি

English	বাংলা অর্থ	English	বাংলা অর্থ
Remedy	আইনি প্রতিকার	Monetary damages	আর্থিক ক্ষতিপূরণ
Guaranty	পরবর্তী ত্রুটি সংশোধনের গ্যারান্টি	Workmanship	কাজের কারিগরি মান
Proportionately	আনুপাতিকভাবে	Contract price	চুক্তিমূল্য

14 Term Final Practice Questions

A. Short questions (3-5 marks)

1. Define liquidated damages.
2. What is a penalty clause?
3. State two purposes of an LD clause.
4. Explain the capping effect of LD.
5. What is a genuine pre-estimate of damage?
6. When may a single LD sum for sectional work fail?
7. What is a condition precedent to LD?
8. Explain time at large.
9. State the JCT procedure before LD deduction.
10. What is the NEC3 position on non-completion certification?
11. Define quantum meruit.
12. What is due expedition?
13. State four effects of completion.
14. Distinguish warranty and guaranty.

B. Broad questions (8-12 marks)

1. Distinguish liquidated damages and penalty damages with examples.
2. Discuss the purpose, effect and enforceability of an LD clause.
3. Explain the defences against enforcement of liquidated damages.
4. Compare the LD procedures under JCT and NEC3 and explain the effect of partial possession.
5. Discuss quantum meruit claims in construction contracts.
6. Explain commencement, progress, completion and the effects of completion.
7. Discuss adjustment of time and the relationship between EOT, employer prevention and time at large.
8. Explain warranty, remedy and guaranty in construction contracts.

Answer-writing formula: Definition -> Contract rule -> Purpose/procedure -> Example or clause -> Effect/defence -> Conclusion

15 Self-Test and Final Revision

1. The primary purpose of LD is:

A. Punishment B. Compensation fixed in advance C. Criminal sanction

2. A penalty is generally:

A. A genuine pre-estimate B. Higher than actual loss and punitive C. Always enforceable

3. JCT SBC 11 cl. 2.32 requires:

A. Oral warning only B. Non-completion certificate and written notice C. No procedure

4. When time is at large:

A. LD remains automatic B. Contractor finishes within reasonable time and LD is inoperable C. Work ends immediately

5. NEC3 X7:

A. Specifically requires engineer certification B. Does not specifically require it C. Prohibits LD

6. Quantum meruit means:

A. Punitive payment B. Reasonable value of work done C. Retention money

7. Practical completion usually:

A. Starts LD B. Stops late-completion LD C. Cancels all warranties

8. A guaranty normally operates:

A. Only before commencement B. After completion for a stated period C. Only during tendering

Answer key: 1-B, 2-B, 3-B, 4-B, 5-B, 6-B, 7-B, 8-B

16 Final 30-Minute Revision Plan

1. 5 minutes: revise C-P, C-C-L, G-P-E and P-C-T-S.

2. 5 minutes: write Dunlop and the JCT/NEC/FIDIC clause list from memory.

3. 5 minutes: revise quantum meruit situations.

4. 5 minutes: revise commencement, progress, completion and P-I-L-R.

5. 5 minutes: practise one LD calculation and one time-at-large explanation.

6. 5 minutes: scan the Bangla vocabulary and warranty-remedy-guaranty table.

Final memory sentence: Agree loss in advance, preserve the completion date through EOT, follow certificates and notices, reduce LD for partial possession, and distinguish reasonable payment from punishment.

Source note: This guide is based only on the uploaded 40-page BECM 3107 lecture slides, including their stated clauses, examples and terminology. It is an exam-revision aid and does not add independent legal advice.